

RESEARCH ARTICLE

KSSD2025: A New Annotated Dataset for Automatic Kidney Stone Segmentation and Evaluation With Modified U-Net Based Deep Learning Models

MURILLO F. BOUZON¹, PAULO H. S. DE SANTANA¹, GABRIEL N. MISSIMA¹,
WEVERSON S. PEREIRA¹, FERNANDO P. RIVERA², GILSON A. GIRALDI³,
OSCAR E. H. FUGITA⁴, AND PAULO S. S. SILVA¹

¹Centro Universitário da FEI, São Paulo 09850-901, Brazil

²Universidad de Huánuco, Huánuco 10000, Peru

³National Laboratory for Scientific Computing, Rio de Janeiro 25651-075, Brazil

⁴Hospital Universitário da Universidade de São Paulo, São Paulo 05508-000, Brazil

Corresponding author: Murillo F. Bouzon (mfbouzon@fei.edu.br)

ABSTRACT Kidney stone disease is a prevalent condition associated with an increased risk of chronic kidney failure. Non-contrast Computed Tomography (CT) remains the gold standard for diagnosis due to its high sensitivity and ability to evaluate clinically relevant stone parameters. However, manual segmentation of stones in CT images is time-consuming and subject to inter-observer variability, highlighting the need for automated solutions. This paper introduces the KSSD2025, a novel dataset comprising 838 manually annotated axial CT images for kidney stone segmentation. We conducted a comprehensive evaluation of four segmentation models based on modified U-Net architectures: U-Net, U-Net++, U-Net3+, and TransU-Net. We evaluated their performance through 5-fold cross-validation and external tests. All models achieved mean Dice scores above 95%, with modified U-Net reaching 97.06% during cross-validation and modified U-Net3+ showing better generalization to external data. Our results confirm that U-Net-based architectures, when trained on high-quality annotations, can achieve accurate and reliable kidney stone segmentation. Our findings suggest that the KSSD2025 dataset offers a valuable resource to advance research in kidney stone analysis and the deployment of AI-assisted diagnostic tools in clinical practice.

INDEX TERMS Kidney stone segmentation, deep learning, U-Net, annotated CT scans, medical imaging.

I. INTRODUCTION

The presence of mineral accumulations in the renal system, known as kidney stones, significantly increases the risk of developing chronic kidney disease, a growing global public health concern [1], [2]. As kidney stone disease has

become increasingly prevalent worldwide, more efficient diagnostic and preventive strategies are needed to mitigate complications and reduce healthcare burdens [3].

Non-contrast Computed Tomography (CT) is currently the gold standard for kidney stone diagnosis, due to its high sensitivity and specificity [4], [5]. In addition to confirming the presence of stones, CT scans allow a detailed examination of clinical features, such as stone size, volume, location,

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and potential composition, which are essential parameters to guide appropriate treatment decisions [6], [7], [8].

However, this diagnostic process is largely manual and time-consuming, requiring radiologists to analyze numerous CT scan slices. Manual delineation of stone regions is particularly labor-intensive and prone to inter-observer variability, leading to inconsistent evaluations and limited treatment planning. The situation is further aggravated by the limited availability of experienced radiologists, especially in low-resource settings. In this context, automating kidney stone segmentation could substantially accelerate the diagnostic workflow, reduce diagnostic bias, and ensure a more consistent analysis.

In recent years, deep learning has emerged as a promising tool in medical imaging, allowing significant advances in automated diagnosis [9] and medical image segmentation [10]. Several studies have explored its application in detecting kidney stones [11], predicting treatment outcomes [12], and estimating stone composition [13]. Despite these advances, a major problem persists: the scarcity of manually annotated datasets for training and validating segmentation models. This limitation arises from the difficulty of acquiring high-quality labeled data, given the expertise required and the time associated with manual annotation.

The detection and segmentation of kidney stones are fundamental tasks in the development of an intelligent kidney stone diagnosis system. Detection involves identifying the presence of stones in imaging exams, which assists in disease diagnosis and surgical preparation. In segmentation, the regions containing the stones are highlighted to provide their precise locations, crucial for planning surgical intervention treatments.

Earlier approaches relied on classical image processing techniques. Thein et al. [14] proposed an automated method for detecting kidney stones on CT scans, utilizing Otsu thresholding, morphological operations, and volumetric analysis. Their approach achieved sensitivity values of 47%, 64%, and 70% in different threshold variations. Jain et al. [15] introduced a non-neural network-based approach for detection and segmentation, employing threshold segmentation, the watershed algorithm, and image processing techniques for noise reduction. Tested on over 150 images from the Kaggle Kidney Stone Images dataset, this approach achieved an SSIM of 0.863 and a PSNR of 30.637.

With advances in deep learning, Convolutional Neural Networks (CNNs) [16] have become a popular solution for medical image analysis, including kidney stone diagnosis. Yildirim et al. [17] trained an XResNet-50 network on 1,799 2D CT images and achieved 96.82% accuracy. Neha [18] combined machine learning techniques for kidney localization and stone segmentation, achieving an Intersection over Union (IoU) of 76.79%.

Among deep learning techniques, the U-Net [19] has become widely used for segmentation tasks, demonstrating its effectiveness in various medical applications [20], [21],

[22], [23], and several studies have applied it for kidney stone segmentation in CT images. Cui et al. [24] developed a U-Net-based model for kidney and stone segmentation, achieving a sensitivity of 95.9% and a Positive Predictive Value (PPV) of 98.7%. Li et al. [25] explored five variations of deep segmentation networks, with the Res U-Net model achieving a Dice score of 96.54%. Kim et al. [26] used U-Net++ for urinary stone segmentation on NCCT scans, reporting a Dice coefficient of 87.56%. Preedanant et al. [27] developed a cascaded U-Net pipeline for segmentation on abdominal X-rays, achieving an F2 score of 71.28%. Rabby et al. [28] applied U-Net for kidney segmentation and YOLOv5/YOLOv7 for stone detection on CT images, where YOLOv7 achieved an accuracy of 99.5% and an IoU of 91.4%. Jang et al. [29] proposed a modified U-Net for urinary system segmentation, achieving a Dice coefficient of 0.8673. In addition to CNN-based approaches, researchers have also explored hybrid models. Elton et al. [30] combined a 3D U-Net for kidney segmentation with a 13-layer CNN classifier, achieving a sensitivity of 0.86 and an area under the curve (AUC) of 0.95. Huang et al. [31] combined Inception-ResNetV2 for classification with U-Net (ResNet34) for segmentation, achieving 99.7% accuracy and 93.7% intersection over union (IoU). Huang et al. [32] introduced a V-Net enhanced by hard example weighting for dual-energy CT segmentation, achieving an AUC of 0.96. Kumar et al. [33] used level set segmentation and Kernel-ELM classification for ultrasound images, obtaining an accuracy of 90.73%.

Although U-Net-based architectures have dominated medical image segmentation, recent studies have explored Transformer-based models to capture long-range dependencies and improve segmentation accuracy in different clinical diseases. Chen [34] introduced TransU-Net, a hybrid CNN-Transformer model that combines U-Net's localization ability with Transformers' global context modeling. Their approach achieved a Dice score of 88.39% for multi-organ segmentation and 67.67% for tumor segmentation. Similarly, Li et al. [35] proposed MultiIB-TransU-Net, integrating information bottleneck blocks with a single Transformer layer, reducing overfitting while maintaining high performance (Dice Score: 81.83% on the Synapse dataset). Lv et al. [36] introduced ECM-TransU-Net, an Edge-enhanced Multi-scale Attention and Convolutional Mamba Transformer U-Net. This model integrates edge detection and multi-scale attention mechanisms to enhance feature extraction and spatial consistency. Evaluated on large-scale CT and MRI datasets, ECM-TransUNet achieved a DSC of 84.68%, with a computational cost of 50.68G FLOPs and 66.47M parameters. Ji et al. [37] proposed BGRD-TransUNet for breast lesion segmentation, introducing novel multi-scale residual feature modules and boundary attention mechanisms, significantly improving IoU and DSC scores over TransUNet.

Other hybrid and non-convolutional approaches have also been investigated. For instance, LSTM-based models have

been proposed to improve prediction based on clinical or imaging sequences. Zhu et al. [38] developed an adaptive feature-weighted LSTM model using routine blood and urine tests to predict the early presence of kidney stones, achieving 89.5% accuracy and an AUC of 0.95. In another work, Jadhav et al. [39] proposed a CNN-LSTM hybrid model trained on ultrasound images for kidney stone detection, reporting an accuracy of 97% and strong generalization capabilities in clinical scenarios. Additionally, Choi et al. [40] introduced PFSH-Net, a parallel frequency-spatial hybrid network that integrates spatial and frequency-domain features for kidney stone segmentation in veterinary CT images, achieving a Dice score of 0.7667.

While deep learning-based models, particularly U-Net variants, have demonstrated success in kidney stone segmentation, a major challenge remains: the scarcity of publicly available annotated datasets. This limitation affects model generalization and clinical applicability. Although techniques such as GAN-based data augmentation [41] and radiomics [42] have been explored, they do not fully address the need for high-quality annotated segmentation datasets.

To address this gap, we introduce the KSSD2025 (Kidney Stone Segmentation Dataset 2025), a novel dataset consisting of 838 CT images with annotated kidney stone segmentation masks. We evaluated its applicability through a comparative study involving four U-Net-based segmentation architectures: U-Net, U-Net++, U-Net3+, and TransU-Net. Our experiments demonstrate that this dataset provides a valuable resource for advancing research in automatic kidney stone segmentation.

The main contributions of our work are the following:

- A novel dataset for kidney stone segmentation, named KSSD2025, comprising 838 manually annotated CT images to train deep learning segmentation models.
- A comparative analysis of modified U-Net, U-Net++, U-Net3+, and TransU-Net architectures for kidney stone segmentation using the KSSD2025.
- A kidney stone segmentation model based on U-Net, achieving a Dice score of 97.06%.

II. MATERIALS AND METHODS

In this work, we propose a novel dataset, named KSSD2025 (Kidney Stone Segmentation Dataset 2025), designed for kidney stone segmentation on axial CT images using deep learning. The proposed methodology is presented in Fig. 1. The pipeline begins with axial CT images containing kidney stones, followed by image pre-processing and manual annotation of ground truth segmentation masks to construct the dataset. We then trained four U-Net-based models using the constructed dataset. The training setup and hyperparameters, along with the segmentation metrics used for evaluation, are presented, enabling an in-depth investigation of their effectiveness for kidney stone segmentation.

The following sections provide a detailed description of each step in the proposed methodology.

A. KIDNEY STONE CT IMAGES

The images used in this study were introduced by Islam et al. [43]. They collected the exams from Dhaka Central International Medical College and Hospital (DCIMCH), Bangladesh, with the hospital authorities' ethical approval. Patients diagnosed with kidney stones, cysts, and tumors voluntarily participated in the study, providing informed consent for data collection.

The dataset contains coronal, sagittal, and axial CT scans from contrast-enhanced and non-contrast studies, acquired following standardized imaging protocols for whole-abdomen scans. To ensure patient privacy, identifying information and metadata were removed before converting the DICOM images to a lossless JPEG format. A total of 12446 images were extracted, comprising 3709 cyst samples, 2283 tumor samples, 1377 kidney stone samples, and 5077 normal samples. This dataset does not contain any segmentation mask annotation.

The original dataset is publicly available on Kaggle,¹ providing a valuable resource for further research on kidney disease diagnosis and medical image analysis.

B. DATASET ANNOTATION

Training precise deep learning models for medical image segmentation heavily relies on high-quality datasets with ground-truth annotations. However, obtaining such datasets is particularly challenging in medical imaging due to the complexity of manual annotations and the limited availability of expert-labeled data. Segmentation of kidney stones on CT scans is especially difficult due to variations in stone size, shape, and contrast levels. To address this issue and enable the training of robust segmentation models, we developed the KSSD2025 dataset, where we added segmentation masks manually annotated from kidney stone regions based on the dataset previously presented by Islam et al. [43].

From the 12446 images available in the original dataset, we selected 838 images that specifically contain kidney stones in axial view. The axial perspective was chosen because it provided the largest number of samples and offered a more suitable anatomical reference to locate and estimate kidney stone sizes during manual annotation. To prepare the dataset for segmentation tasks, we applied a structured pre-processing pipeline to generate ground truth masks:

- **Thresholding:** We empirically set a pixel intensity threshold of 150 to retain only the brightest regions in the image, typically corresponding to high-density structures.
- **Connected Components:** We applied the connected component algorithm to segment different regions in the image and compute their sizes. Regions larger than 300 pixels were discarded, as no kidney stone in the dataset exceeded this size. This step effectively removed

¹<https://www.kaggle.com/datasets/nazmul0087/ct-kidney-dataset-normal-cyst-tumor-and-stone>

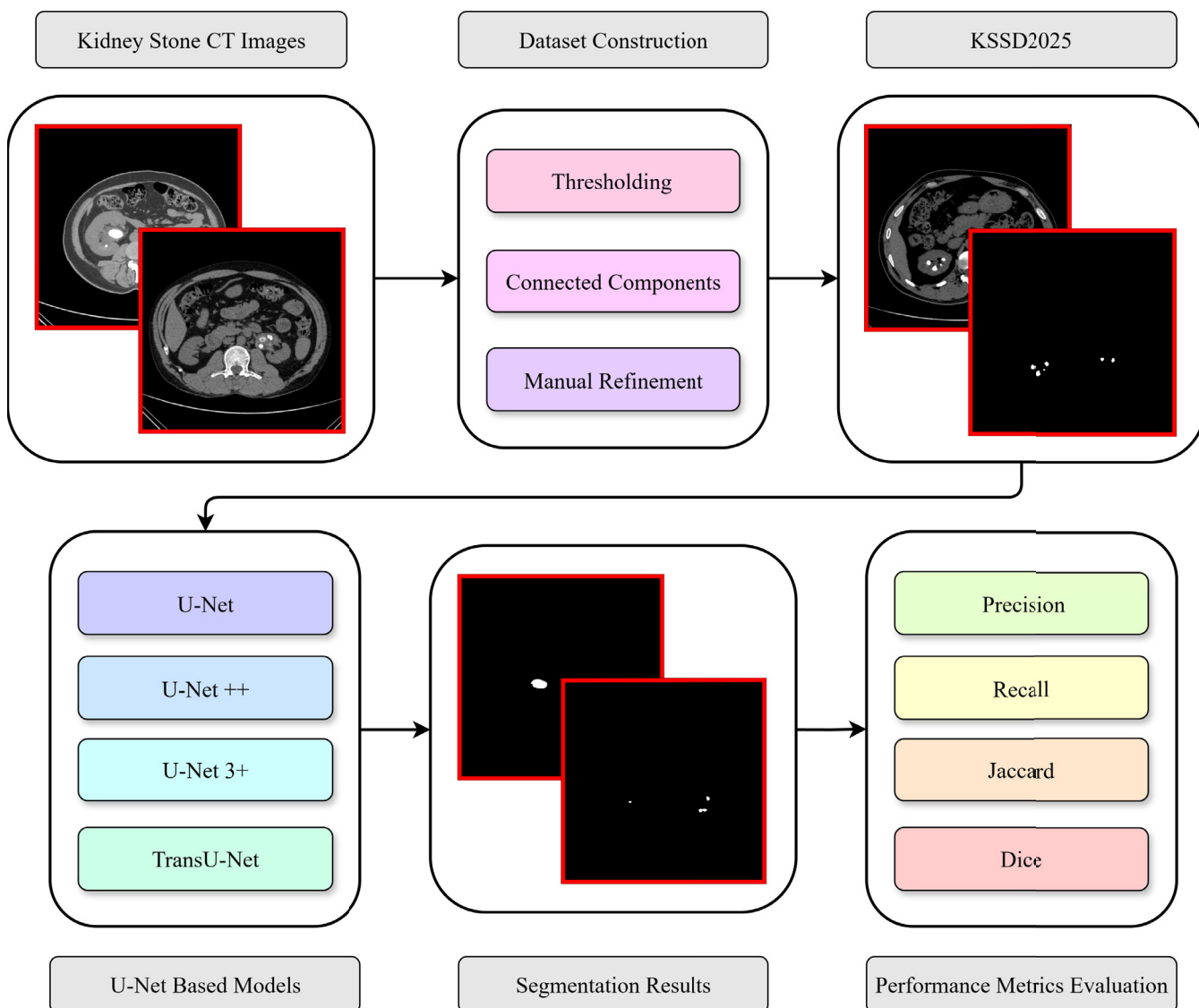


FIGURE 1. Diagram of the proposed KSSD2025 construction and U-Net based models validation.

large bone structures and the table region from the images.

- **Manual Refinement:** A final manual refinement step was performed to eliminate remaining noise and non-stone regions, ensuring precise segmentation. This process was conducted in collaboration with urology and radiology specialists to enhance annotation accuracy.

The ground-truth segmentation masks were systematically paired with their corresponding original images to form a structured dataset after the pre-processing and annotation steps. Each annotated mask precisely delineates the kidney stone regions, ensuring a well-organized dataset that is suitable for training, validation, and testing in deep learning applications. The dataset annotation pipeline is illustrated in Fig. 2. To assess inter-annotator consistency, a second human annotator independently labeled a subset of 100 CT

slices. We then compared these annotations with the original manual labels created with expert support. The average Dice similarity coefficient between the two sets of masks was 74.53%. Representative examples of this comparison are shown in Figure 3. When compared to fully manual annotation, an annotator labeled approximately 100 images per hour using the dataset annotation pipeline, versus only 10 images per hour without it, resulting in a 10× gain in productivity. This efficiency gain highlights the importance of the pre-processing steps in accelerating the labeling process, making large-scale annotation feasible without compromising quality.

Given the scarcity of kidney stone segmentation datasets with expert-labeled ground truth, the development of KSSD2025 represents a significant contribution to the field, providing a high-quality dataset for deep learning models.

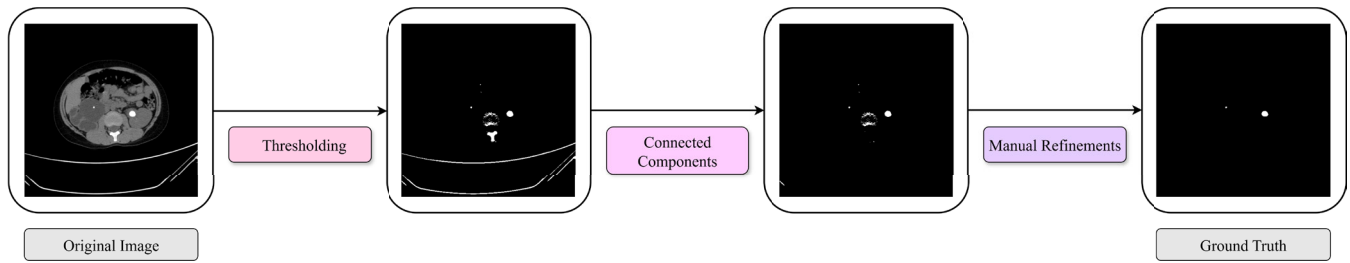


FIGURE 2. Illustration of the dataset annotation pipeline.

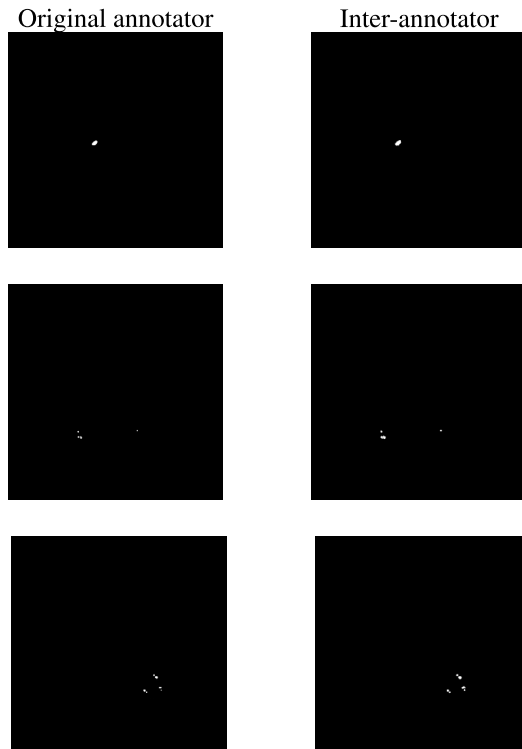


FIGURE 3. Annotated masks comparison samples between annotators.

The original CT images used to construct KSSD2025 were obtained from a public dataset introduced by Islam et al. (2022). Although the dataset's Kaggle page did not specify an open license, we contacted the original authors, who explicitly authorized the use, modification, and redistribution of the data, provided that their work is properly cited. Our annotated dataset is shared with full attribution to the original authors, and detailed licensing information are available in the dataset repository.

The KSSD2025 dataset differs from existing kidney stone segmentation datasets in the following aspects:

- Composed exclusively of axial CT slices following PACS standards used in clinical environments.
- The annotation process combined classical computer vision techniques (thresholding and connected components) as a pre-filter to assist human annotators

- Manual refinement conducted with support from experienced radiologists and urologists.

KSSD2025 is available on Kaggle.²

C. MODIFIED U-NET MODELS

The models trained for kidney stone segmentation in this study are based on the U-Net architecture [19], which has demonstrated remarkable performance in medical image segmentation. Originally proposed for biomedical image processing, the U-Net is widely adopted due to its ability to efficiently learn spatial features and accurately segment small structures in medical images. Although alternative architectures, such as transformers, recurrent models, and hybrid frequency-domain approaches, have gained attention, we opted to evaluate the KSSD2025 using only U-Net-based models due to their optimal performance and widespread clinical adoption.

The original U-Net architecture follows an encoder-decoder structure with a symmetric U-shaped design, as illustrated in Fig. 4. The encoder path, also known as the contracting path, progressively extracts hierarchical features using convolutional layers with ReLU activation, followed by max-pooling layers that reduce spatial dimensions while increasing the number of feature channels. The decoder path, also referred to as the expanding path, reconstructs the segmented regions by upsampling the feature maps, applying transposed convolutions, and concatenating them with corresponding feature maps from the encoder via skip connections.

To improve segmentation results for kidney stone segmentation in KSSD2025, we proposed a modification to the U-Net architecture, adjusting its three main components to a lighter architecture, as follows:

- **Encoder:** The encoder extracts spatial features progressively using convolutional layers (3×3 kernels) with Batch Normalization and ReLU activation. To improve generalization, Dropout layers (0.1–0.3) are applied to improve generalization, and max pooling (2×2 stride) is used for downsampling. The encoder follows a structured hierarchy with feature map sizes of 16, 32, 64, and 128 filters.

²<https://kaggle.com/datasets/72b50f9e4b665a1b76b4d972a7f820b637b69622879be3c18cb51ac530ec2bfc>

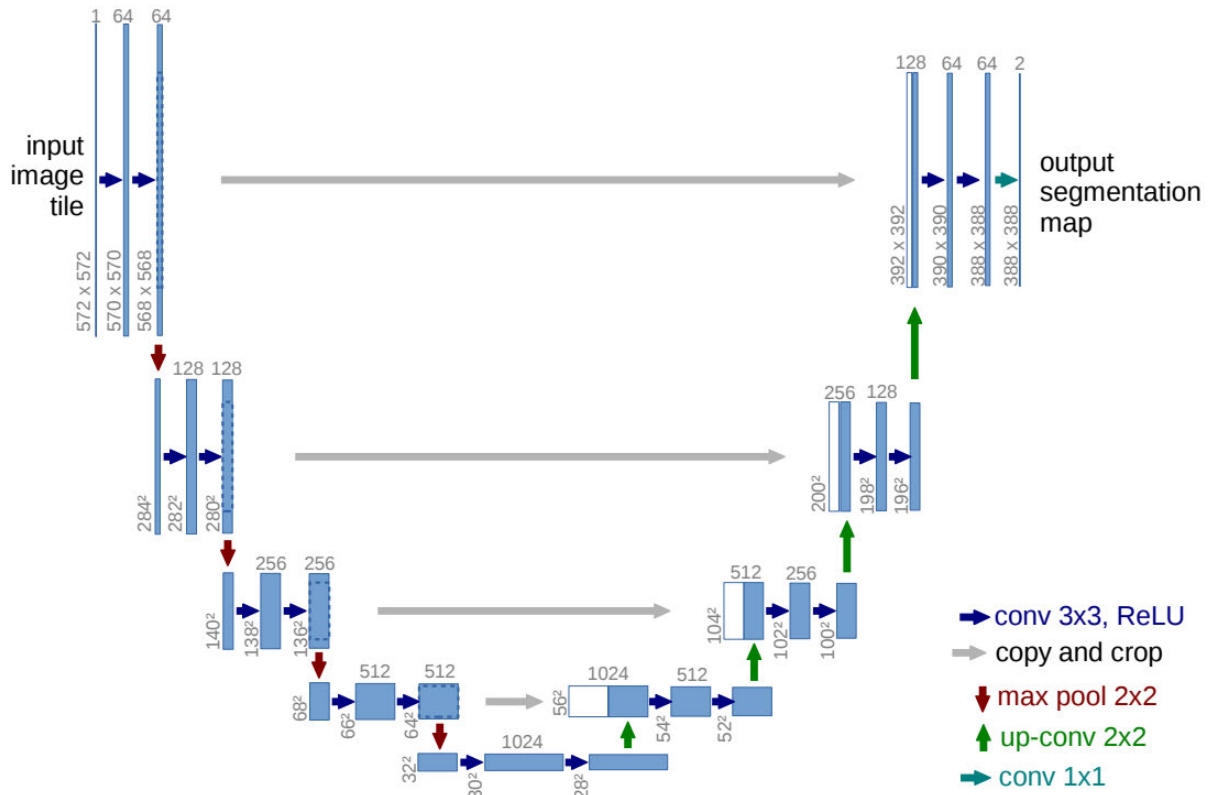


FIGURE 4. Original U-Net architecture.

- Bottleneck: Serving as a bridge between the encoder and decoder, it processes the extracted features with 256 filters to capture high-level representations.
- Decoder: This component uses transposed convolutions (2×2 , stride 2×2) to upsample the feature maps, progressively reconstructing the segmentation mask while incorporating spatial information through skip connections. The decoder reduces the feature map dimensions from 128 to 64, 32, and 16 filters to generate a refined output. The final layer applies a 1×1 convolution with a sigmoid activation function, producing a binary mask representing kidney stone segmentation.

The modified U-Net used in this work is illustrated in Fig. 5. It improves efficiency and performance compared to the original design. Incorporating Batch Normalization (BN) after convolutional layers and strategically placing Dropout layers within the contracting path enhances regularization, helping to mitigate overfitting. Reducing the initial number of filters from 64 to 16 significantly reduces computational cost and memory usage, making the model more suitable for deployment in limited environments. However, the simplified decoder structure may reduce segmentation precision. The reduced filter capacity in the early layers may also limit the extraction of low-level spatial features critical for boundary region segmentation. These trade-offs position the model as a lightweight alternative suited for less complex segmentation

TABLE 1. Comparison between the modified U-Net and the original U-Net in terms of computational cost and resource usage.

Model	Initial filter size	Trainable Parameters	Size
Modified U-Net	16	1.4M	5.3 MB
Original U-Net	64	22.3M	85.2 MB

tasks. Table 1 shows a resource usage comparison between the proposed modified U-Net and the original.

To evaluate the robustness of different U-Net-based approaches, we trained and analyzed the performance of four segmentation models, each incorporating the proposed modifications to the standard U-Net structure. All architectures retain the fundamental encoder-bottleneck-decoder modified framework, applying skip connections for feature fusion and spatial information preservation. The U-Net architectures used are presented as follows:

- U-Net: The baseline model, following the standard U-Net design with an encoder-decoder structure and skip connections to retain spatial information.
- U-Net++ [44]: An enhanced version of U-Net that introduces dense skip pathways between the encoder and decoder, refining feature fusion by incorporating nested convolutional blocks between encoder-decoder pairs, instead of direct skip connections.

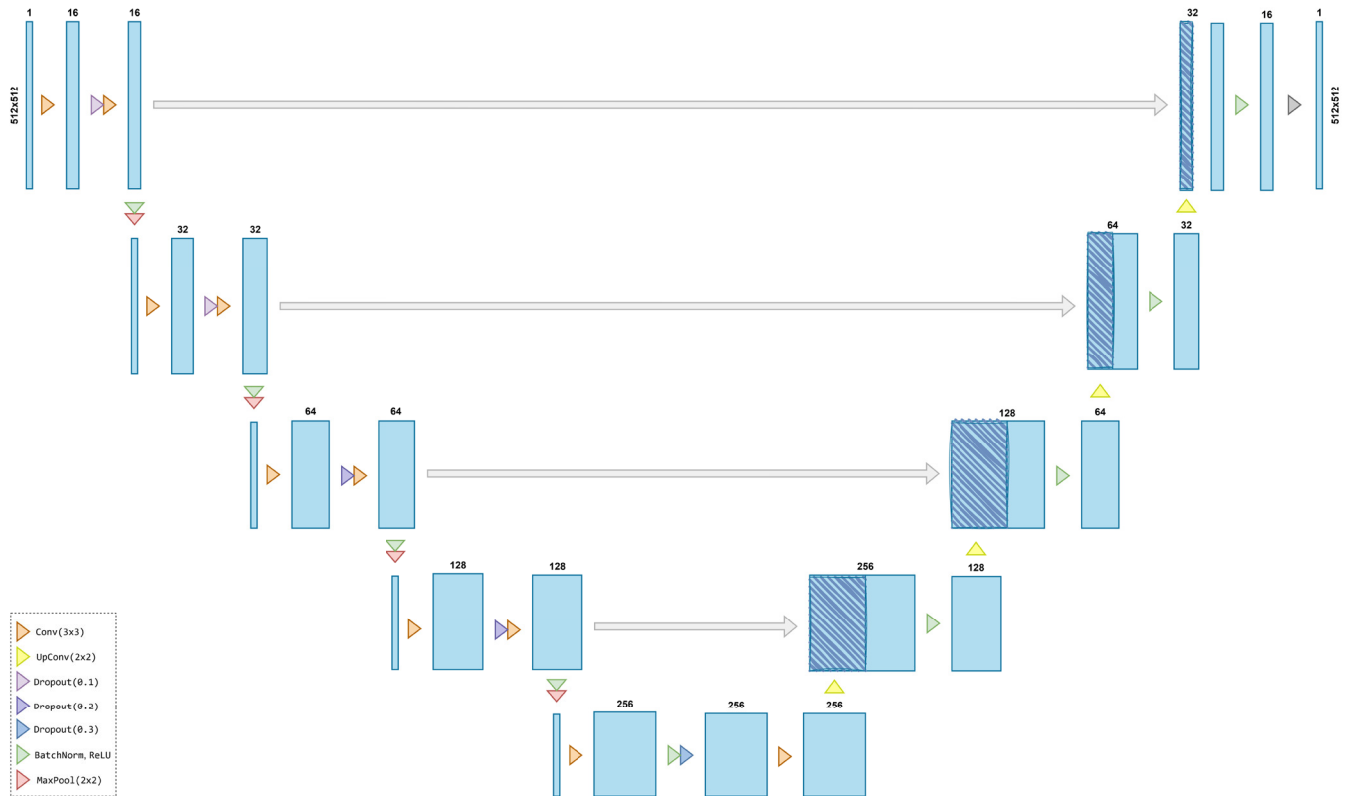


FIGURE 5. Modified U-Net architecture.

- U-Net3+ [45]: An extension that integrates full-scale skip connections, enabling the decoder to aggregate multi-scale information from all encoder levels simultaneously, thereby constructing an enhanced feature representation by merging outputs from multiple resolutions.
- TransU-Net [46]: A hybrid model that incorporates Vision Transformers (ViTs) within the encoder to introduce self-attention mechanisms. The transformer blocks replace part of the traditional convolutional layers, allowing TransU-Net to capture long-range dependencies and enhance segmentation accuracy in complex medical images.

Each of these models was trained using the KSSD2025 dataset, and their performance was evaluated using standard segmentation metrics. The following sections provide details of the evaluation metrics and experimental setup.

D. EVALUATION METRICS

To assess the performance of the U-Net-based models, we employed the Precision, Recall, Jaccard, and Dice Score metrics to evaluate their robustness and effectiveness in correctly segmenting kidney stones on CT exams trained with KSSD2025, and to compare them with existing methods.

Precision quantifies the proportion of correct positive predictions relative to all positive predictions produced by a learning model. In image segmentation, this metric

assesses the model's ability to minimize false positives by calculating the segmented kidney stone pixels relative to all pixels predicted as part of a stone by the model. A high precision value indicates that the model effectively distinguishes between actual kidney stones and non-kidney-stone regions, thereby reducing the risk of oversegmentation. Precision is defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

Recall, also known as sensitivity, measures the model's true positive prediction rate. In image segmentation, this metric evaluates the model's ability to capture the complete extent of the kidney stone, representing the proportion of correctly segmented kidney stone pixels compared to all actual stone pixels in the ground truth. A high recall value indicates that the model effectively minimizes false negatives, thus reducing the likelihood of under-segmentation. Recall is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

The Jaccard index, or Intersection over Union (IoU), quantifies the similarity between the predicted segmentation and the ground truth by measuring their overlap. In image segmentation, this metric evaluates how the model correctly identifies kidney stone regions while penalizing false positives and negatives. A higher Jaccard index indicates a more

accurate segmentation, ensuring that the detected kidney stone regions closely match the ground truth image. The Jaccard index is defined as:

$$\text{Jaccard} = \frac{TP}{TP + FP + FN} \quad (3)$$

The Dice score is a widely used metric for evaluating image segmentation models. It represents the harmonic mean between precision and recall, providing a balanced assessment of segmentation performance. In kidney stone segmentation, the Dice Score measures how well the predicted kidney stone regions align with the ground truth, considering both false positives and false negatives. The Dice Score is defined as follows:

$$\text{Dice Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

We used both Dice and Jaccard metrics to provide a more comprehensive evaluation, as their main distinction lies in how they weigh errors. The Dice Score tends to be more sensitive to small segmented areas, making it particularly useful in medical imaging applications. In contrast, the Jaccard Index is stricter, as it heavily penalizes misclassified regions.

E. TRAINING SETUP AND HYPERPARAMETERS

Our experiments were conducted using Jupyter Lab on a computer desktop with an NVIDIA GeForce RTX 3060 GPU, an AMD Ryzen 5 5500 CPU, and 16 GB of RAM. To assess the generalization capability of the models and ensure robustness, we employed a 5-fold cross-validation strategy to evaluate all models' performance on unseen data. To further improve the model's classification performance and generalization capability, we applied data augmentation to each training batch, using the following transformations:

- Rotation: up to 2.5°;
- Width shift: up to 0.75%;
- Height shift: up to 0.75%;
- Noise range: up to 0.75%;
- Zoom: up to 0.75%;
- Horizontal flip: enabled.

Hyperparameters were empirically selected to optimize segmentation performance. The models were trained for 150 epochs with a batch size of 16. We used the Adam optimizer, with a learning rate of 0.001, due to its adaptive learning capabilities. The Dice loss function (Equation (4)) was applied to optimize the segmentation overlap between predictions and ground truth, as it is effective in addressing medical image segmentation. As described previously, the ReLU activation function was used for each convolution layer. Table 2 summarizes the hyperparameters used in the training process.

The four architectures were trained using the same hyperparameters to ensure a fair comparison. We opted not to fine-tune each model individually to prevent biasing the results toward specific configurations. While alternative

TABLE 2. Hyperparameters used for training the deep-learning architectures.

Hyperparameters	Details
Epochs	150
Batch size	16
Learning rate	0.001
Optimizer	Adam
Activation function	ReLU
Loss function	Dice

learning-rate schedules could potentially benefit individual models, our goal was to evaluate architectural differences under identical training conditions.

III. RESULTS AND DISCUSSION

In this section, we present the experimental evaluation of the KSSD2025 dataset for kidney stone segmentation on axial CT images. We detail a performance investigation of the U-Net-based models, presenting the segmentation metrics obtained in the 5-fold cross-validation, the visual results on the developed dataset, error case analysis, and performance in unseen samples from other data sources. Additionally, we provide a complexity analysis of the proposed models. To contextualize the effectiveness of our approach, we compare the performance of models trained with KSSD2025 with previous studies that employed similar methodologies for kidney stone segmentation. Finally, we discuss the clinical implications of our findings and outline potential applications for future research.

A. 5-FOLD CROSS-VALIDATION PERFORMANCE ANALYSIS

Following the 5-fold cross-validation procedure, we evaluated the mean precision, recall, Jaccard index, and Dice coefficient for each U-Net-based model trained on the KSSD2025 dataset. The average results obtained in the five folds are summarized in Table 3. Among the evaluated architectures, the modified U-Net achieved the best overall performance, with a Dice coefficient of 97.06% and a Jaccard index of 94.65%, indicating a strong similarity between the predicted and ground-truth segmentation masks. U-Net++ and U-Net3+ also achieved high performance metrics, with Dice scores of 96.63% and 96.68%, respectively. In contrast, TransU-Net achieved the lowest segmentation performance, with a Dice coefficient of 95.53% and a Jaccard index of 92.21%. Furthermore, its higher standard deviations in recall and Jaccard index point to less consistent predictions. Although the attention mechanisms in TransU-Net are designed to improve global context, they may be less effective for detecting small, high-contrast regions such as kidney stones, especially when trained with a limited number of annotated samples.

In summary, the modified U-Net demonstrated superior assertiveness in kidney stone segmentation. The remaining architectures also produced satisfactory results, all achieving Dice scores greater than 95%, which highlights the robustness

TABLE 3. 5-fold cross validation results of U-Net based models trained with KSSD2025.

Model	Precision (%)	Recall (%)	Jaccard (%)	Dice (%)
U-Net	97.83 ± 0.33	96.61 ± 0.77	94.65 ± 0.62	97.06 ± 0.39
U-Net++	97.16 ± 0.68	96.44 ± 0.67	93.97 ± 0.83	96.63 ± 0.63
U-Net3+	96.88 ± 0.34	96.90 ± 1.02	94.02 ± 0.86	96.68 ± 0.61
TransU-Net	96.26 ± 0.52	95.39 ± 1.24	92.21 ± 1.27	95.53 ± 0.89

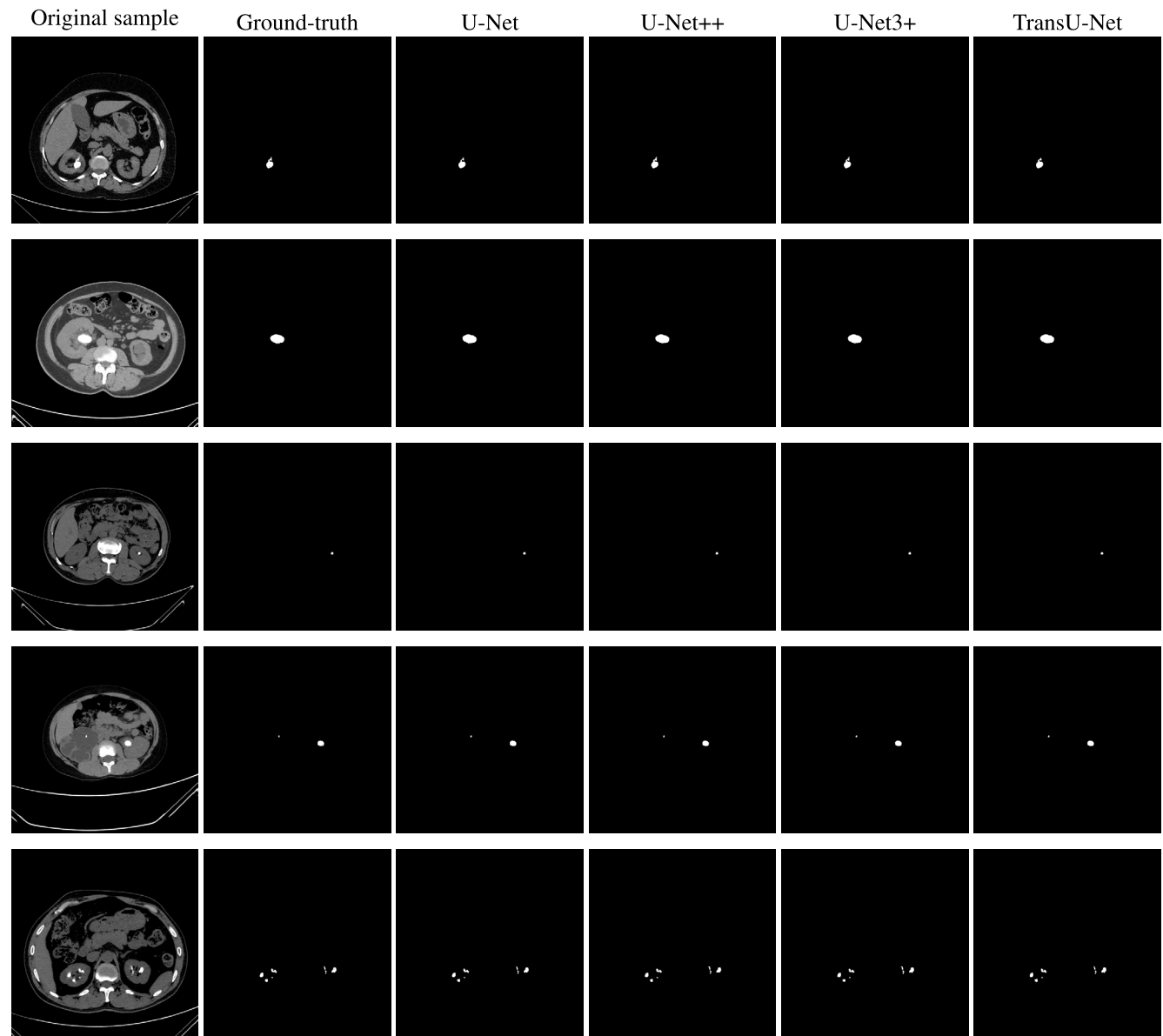


FIGURE 6. Segmentation results from each U-Net based model trained with KSSD2025. The first column shows the original samples. The second column shows the ground-truth sample manually annotated. The third column is the output prediction image obtained by the U-Net model. The fourth column is the output prediction image obtained by the U-Net++ model. The fifth column is the output prediction image obtained by the U-Net3+ model. The sixth column is the output prediction image obtained by the TransU-Net model.

of the KSSD2025 dataset as a training resource for kidney stone segmentation models.

B. VISUAL RESULTS ANALYSIS

To further assess segmentation quality, a visual inspection was performed on representative samples from the validation

set. Fig. 6 illustrates the segmentation results obtained by the U-Net, U-Net++, U-Net3+, and TransU-Net models, including cases with single and multiple kidney stones.

Across the examples shown, all models produced segmentation masks that closely resemble the manually annotated ground-truth, corroborating the high Dice scores reported in

TABLE 4. Mean Dice coefficient, confidence intervals, and normality test for each architecture.

Architecture	Mean Dice	95% CI	Shapiro-Wilk (W/p)
U-Net	0.9706	[0.9651, 0.9761]	0.9704 / 0.8775
U-Net++	0.9663	[0.9575, 0.9751]	0.9842 / 0.9556
U-Net3+	0.9669	[0.9598, 0.9740]	0.9791 / 0.9296
TransU-Net	0.9553	[0.9429, 0.9677]	0.9889 / 0.9757

the quantitative analysis. The selected cases include diverse clinical scenarios, such as kidney stones located in the left or right kidney, varying in size and quantity (single vs. multiple stones).

The U-Net, U-Net++, and U-Net3+ models yielded particularly accurate segmentations in both simple (single large stone) and complex (multiple small stones) cases, demonstrating strong localization capabilities. In contrast, the TransU-Net occasionally produced slightly more diffuse or fragmented predictions, especially in the presence of multiple stones, which may explain its relatively lower Dice performance and higher variance.

Overall, the qualitative results support the effectiveness of all evaluated models for kidney stone segmentation, while highlighting the superior consistency provided by the modified U-Net architecture.

C. STATISTICAL ANALYSIS

To assess the statistical significance of performance differences between the evaluated segmentation models, we conducted a comprehensive statistical analysis based on the Dice coefficient values.

Table 4 presents the mean Dice coefficient, 95% confidence intervals, and results of the Shapiro-Wilk normality test [47] for each architecture. All models exhibited near-normal performance distributions, with Shapiro-Wilk p-values above 0.87, validating the application of parametric tests.

To further compare the models, we performed one-sided paired t-tests ($A > B$) between each pair of architectures. The results are shown in Figure 7, where each matrix entry corresponds to the p-value of the hypothesis that Architecture A significantly outperforms Architecture B.

The analysis indicates that U-Net achieved the highest mean performance and significantly outperformed both U-Net3+ and TransU-Net ($p = 0.02$ in both cases). U-Net++ also significantly outperformed TransU-Net ($p < 0.001$), though its performance was not statistically different from U-Net or U-Net3+. These results support the conclusion that, under our experimental conditions, the classical U-Net architecture achieved the most robust segmentation performance.

D. ERROR CASES ANALYSIS

Despite the high performance observed across all models, some challenging cases led to segmentation errors, as illustrated in Fig. 8. These samples highlight failure cases where

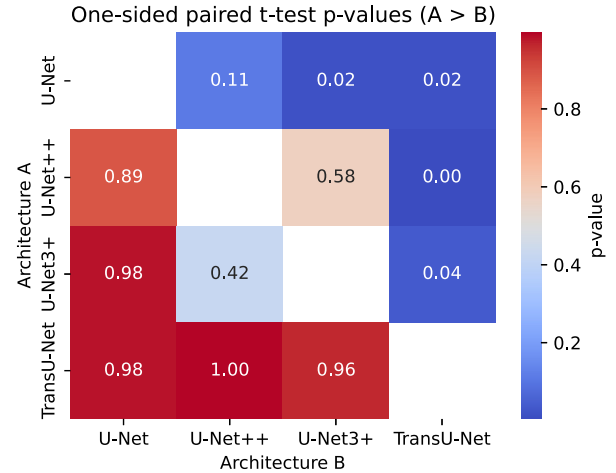


FIGURE 7. One-sided paired t-test p-values ($A > B$). Lower p-values indicate statistically significant superiority of Architecture A over B.

kidney stones were either partially segmented or completely missed by one or more models.

In the examples shown, the ground-truth annotations reveal the presence of smaller stones with low contrast, which can be difficult to distinguish from surrounding tissue. Some models struggled to accurately segment these regions, occasionally generating no prediction signals, as seen in the U-Net, U-Net++ and TransU-Net cases. Such errors are particularly evident when the stones are close to heterogeneous textures. Interestingly, although the TransU-Net model achieved the lowest average performance metrics across the cross-validation results, it was the model that most accurately approximated the ground-truth segmentation in the example of the last row of Fig. 8. This result suggests that TransU-Net may have a higher capacity for generalization in complex scenarios, despite its lower overall performance on the KSSD2025 cross-validation.

These error cases emphasize the limitations of the models when dealing with subtle targets in challenging conditions. Addressing these issues could involve incorporating attention mechanisms fitted to small object detection, or augmenting the dataset with more representative difficult cases.

E. MODEL ASSESSMENT ON EXTERNAL DATA

To further evaluate the generalization of the models, we conducted a supplementary test using samples from an external dataset collected by radiologists from the Hospital Universitario da Universidade de São Paulo, not included in the KSSD2025 training or validation process. This external dataset contained samples divided by anatomical parts of the urinary system where the stones were located. Table 5 summarizes the Dice scores obtained for each anatomical location, specifically the right kidney (5 samples), left kidney (3 samples) and ureter (9 samples), totaling 17 samples from the external test set.

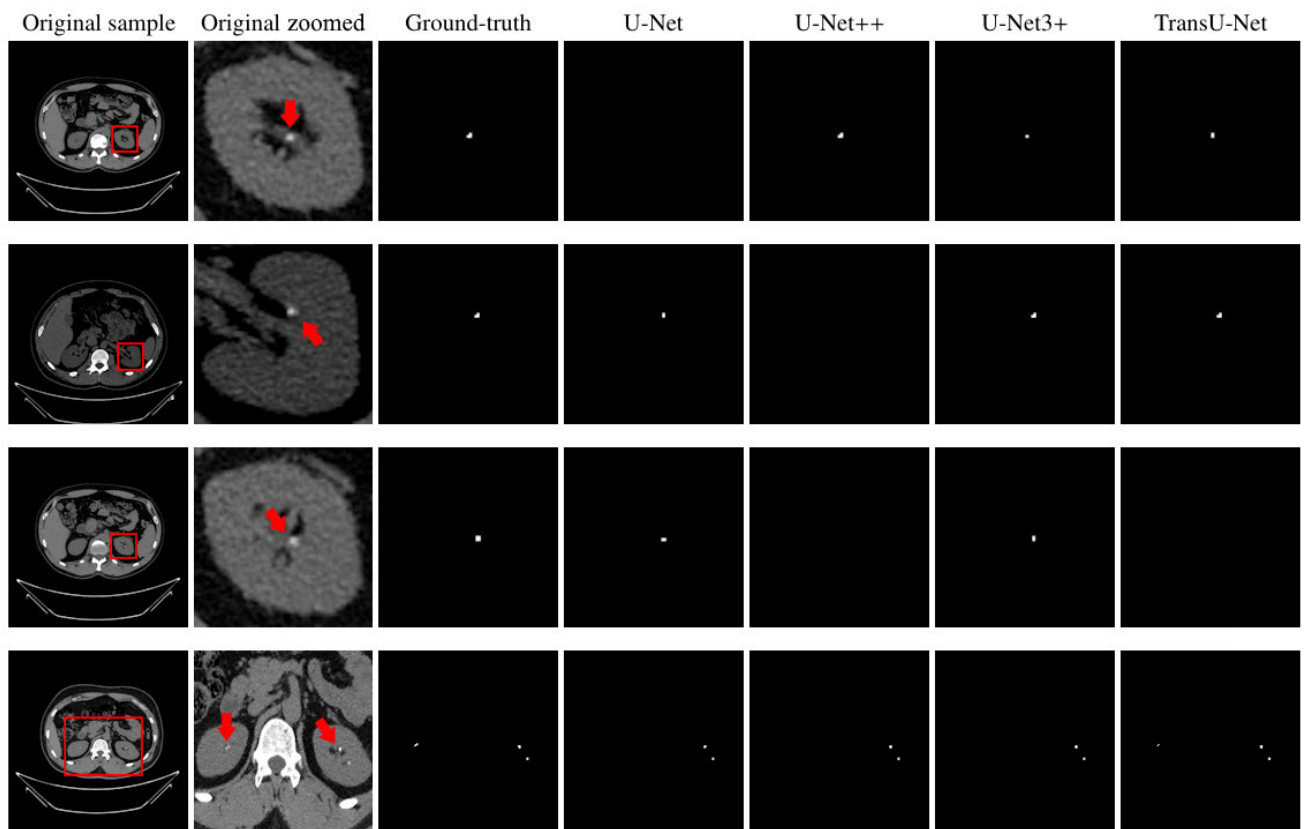


FIGURE 8. Segmentation results revealed that the U-Net based models presented errors. The first column displays the original samples. The second column displays the original samples zoomed. The third column displays the ground-truth sample zoomed. The fourth column shows the output prediction image obtained by the U-Net model. The fifth column shows the output prediction image obtained by the U-Net++ model. The sixth column shows the output prediction image obtained by the U-Net3+ model. The seventh column shows the output prediction image obtained by the TransU-Net model.

TABLE 5. Dice coefficients (%) obtained by each model when tested on an external dataset, segmented by anatomical region.

Model	Right Kidney Stone (5 samples)	Left Kidney Stone (3 samples)	Ureter Stone (9 samples)	Average (17 samples)
U-Net	84,05%	81,78%	60,49%	75,44%
U-Net++	84,05%	83,45%	73,94%	80,48%
U-Net3+	84,90%	83,45%	91,99%	86,78%
TransU-Net	84,05%	84,67%	72,57%	80,45%

The results reveal a noticeable performance drop compared to the KSSD2025 cross-validation (Section III-A), particularly in the segmentation of ureter stones. This decline is expected due to possible variations in image acquisition parameters, contrast levels, and patient anatomy between the datasets.

Among the evaluated models, U-Net3+ achieved the highest average Dice score (86.78%). U-Net, U-Net++, and TransU-Net demonstrated slightly lower performance, especially in detecting ureter stones, as seen in Table 5, which are more challenging to detect due to their small size and low contrast.

Fig. 9 illustrates representative segmentation outputs. Visual analysis confirms that most of the segmentation

masks generated by the models remain reasonably close to the ground truth, especially for kidney stones. However, the segmentation of ureter stones remains inconsistent, due to the distinct anatomical location, smaller stone sizes, and more variable positioning within the urinary tract. Unlike kidney stones, which tend to be larger and centered within the kidney regions, ureteral stones often appear in narrow, tortuous areas and can be easily confused with surrounding soft tissues. These anatomical and spatial differences challenge the model's generalization, especially when trained primarily on kidney stone cases. This analysis suggests the need for model enhancements or fine-tuning when dealing with different clinical scenarios.

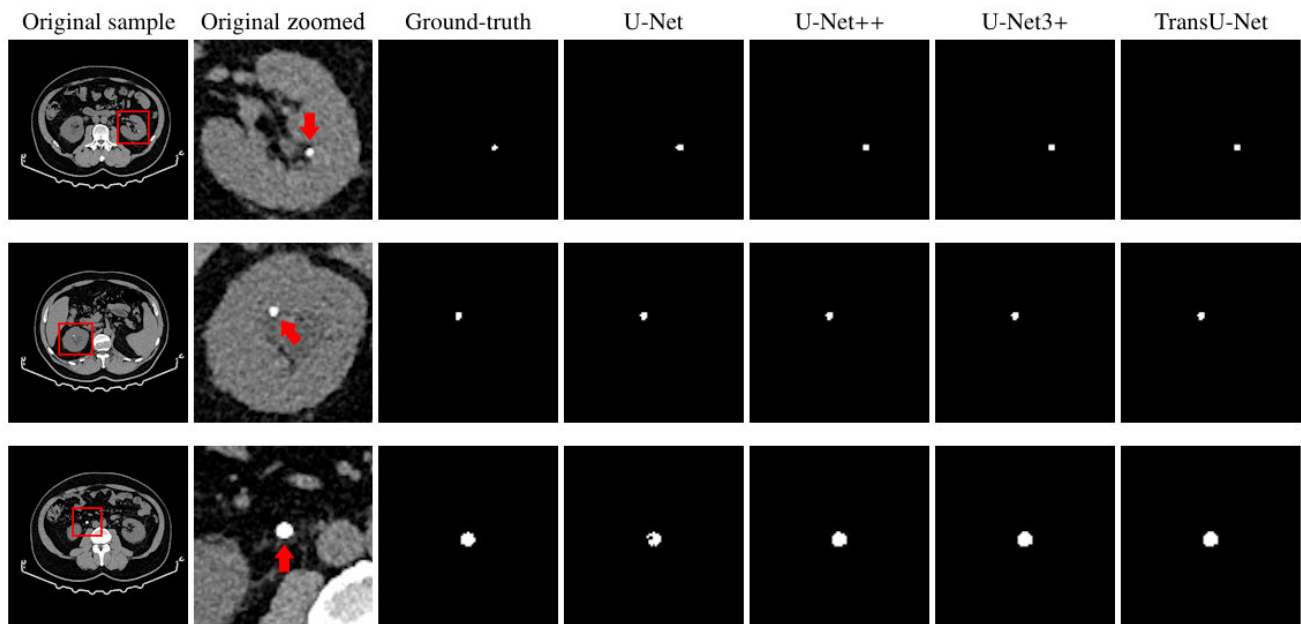


FIGURE 9. Segmentation results on external data samples. The first row is the results obtained in a left kidney stone sample. The second row is the results obtained in a right kidney stone sample. The third row is the results obtained in a ureter stone sample. All models were trained exclusively on the KSSD2025 dataset.

These findings highlight the importance of evaluating models with external data and demonstrate that KSSD2025 trained architectures achieved a promising level of generalization. Further robustness could be achieved with the inclusion of more diverse training samples from other sources.

F. MODEL ASSESSMENT ON SAMPLES WITHOUT KIDNEY STONES

To ensure robustness and verify the ability of the trained models to avoid false positives, we also evaluated their performance on CT samples without kidney stones. These cases are critical in clinical settings, where incorrect segmentation of non-existent stones could lead to unnecessary interventions. Those images were also collected by radiologists from the Hospital Universitario da Universidade de São Paulo and were not included in the KSSD2025 training or validation process. Fig. 10 presents qualitative results on such samples. All four models based on modified U-Net architecture, U-Net, U-Net++, U-Net3+, and TransU-Net, produced empty segmentation masks, correctly identifying the absence of kidney stones. These outputs are consistent with the ground-truth annotations, which do not contain labeled regions. This behavior demonstrates that the models are capable not only of detecting kidney stones when present but also effective in avoiding false positives in healthy kidneys. In this specific evaluation, all models achieved a segmentation accuracy of 100% and a false-positive rate of 0%, perfectly matching the ground-truth masks and confirming their reliability in negative cases. Although the matching was perfect on the

samples tested in this study, further experiments with a large number of non-kidney stone samples are necessary to validate the proposed trained models' effectiveness in avoiding false positives.

G. COMPLEXITY ANALYSIS

Table 6 summarizes the computational complexity of each U-Net based model. As expected, U-Net is the most efficient architecture, requiring the least memory (3.2 GB), having the fewest parameters (1.39M), and the shortest training time. U-Net++ and U-Net3+ introduce more sophisticated skip connections and supervision mechanisms, resulting in moderate increases in complexity. In contrast, TransU-Net, which integrates transformer blocks, is the most computationally demanding, with 4.51M parameters and 18 GB of memory usage.

Despite its simplicity, U-Net achieved the best performance during cross-validation, whereas U-Net3+ performed best on external data. This highlights that, in this binary segmentation task with a limited dataset, simpler convolutional architectures may outperform more complex models. The relatively low difficulty of the task and the small number of samples in the training set likely limited the advantages of deeper and transformer-based models, which typically require larger datasets to reach their full potential.

H. STATE-OF-THE-ART COMPARISON

We compared our results with two recent studies that also addressed the task of kidney stone segmentation [26], [40]. Table 7 summarizes the main differences between these

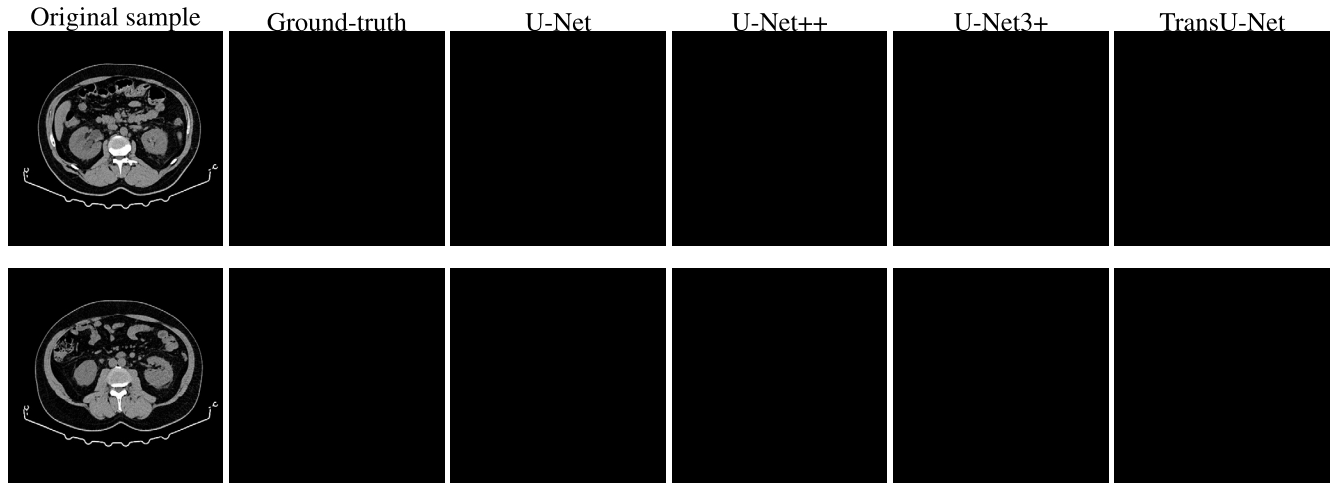


FIGURE 10. Segmentation results on samples without kidney stones. The first column shows the original CT slices. The second column presents the manually annotated ground-truth masks. The third to sixth columns show the predicted segmentation masks generated by the U-Net, U-Net++, U-Net3+, and TransU-Net models, respectively. All models were trained exclusively on the KSSD2025 dataset.

TABLE 6. Complexity information of each U-Net based model used in this work.

Model	Trainable Parameters (M)	Memory Usage (GB)	Average Training Time (s)	Model Size (MB)
U-Net	1.39	3.2	1416	16.5
U-Net++	1.94	8.5	1859	23
U-Net3+	2.31	12	2066	27.3
TransU-Net	4.51	18	3254	53.3

approaches and our work, highlighting key methodological aspects and performance indicators.

Our dataset comprises 838 annotated CT images, over twice than the 410 samples used by Kim et al. [26] and significantly larger than the 63 images used by Choi et al. [40]. This larger dataset likely contributed to the improved segmentation performance observed in our work, as a higher sample quantity typically enhances the generalization of deep learning models.

In terms of network architecture, Kim et al. [26] used U-Net++, whereas Choi et al. [40] employed PFSH-Net, a customized architecture. Our work conducted a broader comparison across four architectures: U-Net, U-Net++, U-Net3+, and TransU-Net. Among these, our modified U-Net, despite being the simplest, achieved the best Dice score during cross-validation (97.06%). In contrast, Kim et al. reported 87.56% for axial slices using U-Net++, and 88.44% when combining axial, coronal, and sagittal views. Choi et al. obtained a lower score of 65.08% with PFSH-Net. Notably, the work of Choi et al. focused on detecting kidney stones in dogs rather than humans. This difference in subject type may introduce additional variability and complexity to the segmentation task due to anatomical and physiological differences, which could explain the considerably lower performance observed in their results.

Interestingly, although Kim et al. employed three CT perspectives and an early stop strategy, their model did not surpass our results, which used only the axial perspective and

no early stopping. This indicates that the axial view alone was sufficient to achieve high performance in the binary segmentation of kidney stones in our dataset. Furthermore, it suggests that model accuracy may depend more on the quality and quantity of the data than on architectural complexity or the use of multiview input.

Regarding training protocols, we used a larger batch size (16 vs. 12 and 2) and trained for more epochs than Kim et al. (100 epochs) and equal to Choi et al. (150 epochs). We also applied data augmentation, unlike Choi et al., which may explain part of the performance gap. Additionally, Kim et al. and Choi et al. used composite loss functions combining Dice and Cross-Entropy terms, whereas we used only the Dice loss.

A notable aspect of Kim et al.'s work is the clinical analysis step, where they compared model outputs to clinical measurements obtained from patient records. While our study did not include such an analysis, we recognize its value and consider it a relevant direction for future work.

In summary, our approach achieved state-of-the-art segmentation performance while relying on a single CT view and relatively simple architectures. Our results can be attributed to using a larger and well-annotated dataset, and to the slight modifications of our proposed U-Net, leading to a lightweight high-performance segmentation model.

I. CLINICAL APPLICATIONS

The automatic segmentation of kidney stones using deep learning techniques represents a valuable advancement in the

TABLE 7. Comparison between our work and state-of-the-art approaches.

Paper	Our work	Kim et al. (2023) [26]	Choi et al. (2025) [40]
Dataset size	838	410	63
Network Architectures	U-Net, U-Net++, U-Net3+, and TransU-Net	U-Net++	PFSH-Net
Batch size	16	12	2
Epochs	150	100	150
Early stop	No	10 epochs	No
Loss Function	Dice	Dice + 0.1CE	0.3Dice + 0.7CE
Cross-validation	5-fold	5-fold	5-fold
Data Augmentation	Yes	Yes	No
CT scan perspective	Axial	Axial, Coronal and Sagittal	Axial
Dice score	97.06%	88.44%(additive model) 87.56% (axial only)	65.08% (Kidney stone)
Clinical Analysis	No	Yes	No
Subject	Human	Human	Dog

clinical workflow for the diagnosis and treatment of kidney diseases. The high segmentation performance achieved by our models, particularly the U-Net and U-Net3+ modified architectures, demonstrates their potential for integration into computer-aided diagnosis (CAD) systems, providing support to radiologists and urologists in analyzing CT scans with increased efficiency and accuracy.

A natural step toward clinical adoption is the deployment of our model into Picture Archiving and Communication Systems (PACS), allowing for real-time processing of CT images and providing clinicians with automated overlays of segmented kidney stones directly within the imaging interface. This automation may significantly reduce the time and cognitive burden of manual annotation, enhance reproducibility, and minimize interobserver variability. However, clinical deployment requires rigorous validation across different hospital environments to ensure model robustness. Variability in patient anatomy, CT scanner models, image acquisition protocols, and noise levels must be considered.

One of the main concerns is the ability of models to generalize across different imaging devices, acquisition protocols, institutions, and patient demographics. Models trained on data from a specific scanner or population may experience a performance drop when applied in new clinical environments due to domain shift. To address this, future work may involve domain adaptation techniques, data harmonization methods, or training with multi-institutional datasets. Additionally, strategies such as transfer learning and federated learning can promote robustness while preserving data privacy. Before deployment, extensive validation in real-world clinical settings is necessary to ensure safety, reliability, and integration into diagnostic workflows.

Although our model has shown high performance on the present dataset, further testing under various clinical conditions is essential to evaluate its generalization and avoid potential diagnostic mistakes.

J. STUDY LIMITATIONS

Despite encouraging results, some limitations must be acknowledged. First, the dataset used in this study was

collected under relatively controlled conditions, with images acquired from a limited number of sources (only from one hospital); this may affect the model's ability to generalize well to CT images acquired using different devices or protocols.

Second, although previous studies have demonstrated the utility of incorporating multiple CT perspectives (axial, coronal, and sagittal) to improve segmentation performance, our model was trained and validated solely using axial images. While our results suggest that the axial view alone can yield highly accurate segmentation for this task, incorporating multi-view information may further enhance robustness and should be investigated in future research.

Third, the current work does not include a clinical correlation between segmentation outputs and relevant patient-level metrics, such as stone burden, Hounsfield density, or clinical outcomes. This omission limits the direct clinical interpretability of our findings and makes it difficult to evaluate them against diagnostic and therapeutic decisions based on ground-truth information.

Furthermore, we acknowledge the absence of an ablation study in this work. Our contribution does not involve the addition of architectural modules, but a minimal modification to U-Net-based models by reducing the filter size in convolutional layers, aiming to explore lighter configurations with lower computational cost. Thus, An ablation analysis focused on isolating the effect of multiple architectural components is not directly applicable in this work. Nevertheless, assessing the isolated impact of this modification across different architectures is a promising avenue for future studies.

Finally, although transformer-based architectures, such as TransUNet, were included, their performance did not surpass that of U-Net-based models; this may be attributed to the relatively small size of our dataset, as transformer models typically require significantly more data to fully leverage their representational capacity.

K. FUTURE DIRECTIONS

Future work will focus on improving the clinical applicability and generalization capacity of the proposed models. A

priority is to expand the dataset with additional CT images from multiple medical centers, including scans obtained using different imaging devices and protocols, to enable domain generalization and improve the robustness of the models in real-world clinical scenarios.

We also aim to investigate the fusion of multi-planar CT perspectives, including axial, coronal, and sagittal perspectives, using either early or late fusion strategies, which could capture more comprehensive spatial features and potentially lead to further improvements in segmentation accuracy, especially for irregularly shaped or small stones.

Another direction is to conduct a clinical validation study in which segmented kidney stones will be compared with manual measurements of stone size, volume, and density, allowing for the automatic extraction of clinically meaningful characteristics that may support physicians in diagnosis, treatment planning, and monitoring disease progression or recurrence.

From a technical standpoint, further exploration of hybrid CNN-transformer architectures could be beneficial, particularly in conjunction with dataset expansion. Additionally, optimizing computational efficiency through model pruning or deployment on edge devices could facilitate real-time applications in low-resource clinical environments.

Lastly, our goal is to bridge the gap between technical performance and clinical utility. To this end, we intend to develop a prototype system integrated with PACS by exporting the segmentation mask as a JSON file containing contour coordinates or as a DICOM object with embedded annotations. These formats can be automatically overlaid into the original CT scans within radiology workstations, enabling enhanced visualization and reducing manual workload for radiologists. This feature can be prospectively evaluated in collaboration with radiology departments, marking a significant step toward the routine clinical adoption of deep learning-based kidney stone segmentation tools.

IV. CONCLUSION

In this work, we introduced the KSSD2025, a novel dataset containing 838 manually annotated axial CT images for kidney stone segmentation. To validate its applicability, we trained and evaluated four different deep learning models: U-Net, U-Net++, U-Net3+, and TransUNet. All architectures achieved high segmentation performance, with average Dice scores exceeding 95%. The U-Net model obtained the best performance during cross-validation, with a mean Dice score of 97.06%, while the U-Net3+ demonstrated superior generalization when evaluated on an external dataset. Although TransUNet was the most complex model, it did not outperform simpler architectures in the performed tests, likely due to the limited dataset size and the binary nature of the segmentation task. Compared to related studies, our approach achieved state-of-the-art performance, even when using only axial slices, whereas other works employed multi-view strategies. These findings suggest that the larger dataset and the careful annotation process were key factors

in enabling high segmentation accuracy. These findings highlight the potential of our method to help automate kidney stone analysis, allowing for the extraction of relevant clinical metrics such as stone size, location, and morphology.

In future work, we plan to increase dataset diversity by including images from different imaging centers and CT machines, thereby improving generalization to unseen data. We also aim to explore the combination of multiple CT perspectives (axial, coronal, and sagittal) and evaluate more advanced architectures. Finally, we intend to integrate clinical analysis to correlate segmentation outcomes with patient-specific metrics and implement our model in real clinical environments, potentially as part of a PACS-integrated decision-support system for kidney disease management.

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MURILLO F. BOUZON received the B.Sc. degree in computer science and the M.Sc. degree from the Centro Universitário da FEI, São Bernardo do Campo, Brazil, where he is currently pursuing the Ph.D. degree in electrical engineering, all with emphasis on signal and image processing. He is also a Researcher with the Centro Universitário da FEI. His research interests include medical image analysis using computer vision techniques, with a focus on kidney stone diagnosis and segmentation in computed tomography images, as well as surgical trajectory planning for ureteroscopy procedures.



intelligent control systems.

PAULO H. S. DE SANTANA received the B.Eng. degree in control and automation engineering from the Centro Universitário das Faculdades Metropolitanas Unidas (FMU), São Paulo, Brazil, the M.B.A. degree in project management from UNICA, and the Technical degree in mechatronics from the Fundação Educacional Inaciana (FEI), São Bernardo do Campo, Brazil. His research interests include computer vision applied to industrial environments, robotic automation, and

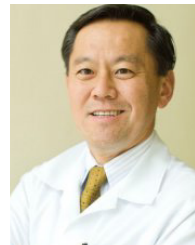


(LNCC), Petrópolis, Brazil. His research interests include image segmentation, deformable models, image processing, statistical learning, and fluid animation.

GILSON A. GIRALDI received the B.Sc. degree in mathematics from the Pontifical Catholic University of Campinas, Campinas, Brazil, in 1986, the M.Sc. degree in applied mathematics from the University of Campinas (UNICAMP), Campinas, in 1993, and the Ph.D. degree in systems and computing engineering from the Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil, in 2000. He is currently a Senior Researcher with the National Laboratory for Scientific Computing



GABRIEL N. MISSIMA is currently pursuing the B.Sc. degree in computer science with the Centro Universitário da FEI, São Bernardo do Campo, Brazil. His academic interests include kidney stone diagnosis using ensemble deep learning models.



OSCAR E. H. FUGITA was born in Brazil. He received the M.D. degree from the School of Medicine, University of São Paulo (USP), São Paulo, Brazil, in 1992, and the Ph.D. degree in urology from the University of São Paulo, in 2001. He completed his Medical Residency in general surgery and urology at Hospital das Clínicas, University of São Paulo, in 1994 and 1997, respectively. From 2000 to 2001, he was a fellow of endourology and videolaparoscopy at the Johns Hopkins University–Brady Urological Institute, Baltimore, MD, USA, under the Endourological Society. He is currently an Associate Professor with the Department of Surgery, School of Medicine, University of São Paulo (USP), and the Director of the Center for Innovation and Technology (INTEC), University Hospital of USP. He is also the Director of the Endourology and Laparoscopy Fellowship Program at the Botucatu Medical School, São Paulo State University (UNESP), the only center accredited by the Endourological Society, Brazil, since 2004. His clinical and research interests include minimally invasive surgery, laparoscopic and robotic urologic surgery, urinary stone disease, and health innovation. He has been a member of the Technical Chamber of Urology at the Regional Medical Council of São Paulo (CREMESP), since 2003. He organized the book *Diretrizes de Laparoscopia Urológica*, which was selected by UNESP Press and nominated for the 64th Jabuti Prize by Brazilian Book Chamber.

He received the M.D. degree from the School of Medicine, University of São Paulo (USP), São Paulo, Brazil, in 1992, and the Ph.D. degree in urology from the University of São Paulo, in 2001. He completed his Medical Residency in general surgery and urology at Hospital das Clínicas, University of São Paulo, in 1994 and 1997, respectively. From 2000 to 2001, he was a fellow of endourology and videolaparoscopy at the Johns Hopkins University–Brady Urological Institute, Baltimore, MD, USA, under the Endourological Society. He is currently an Associate Professor with the Department of Surgery, School of Medicine, University of São Paulo (USP), and the Director of the Center for Innovation and Technology (INTEC), University Hospital of USP. He is also the Director of the Endourology and Laparoscopy Fellowship Program at the Botucatu Medical School, São Paulo State University (UNESP), the only center accredited by the Endourological Society, Brazil, since 2004. His clinical and research interests include minimally invasive surgery, laparoscopic and robotic urologic surgery, urinary stone disease, and health innovation. He has been a member of the Technical Chamber of Urology at the Regional Medical Council of São Paulo (CREMESP), since 2003. He organized the book *Diretrizes de Laparoscopia Urológica*, which was selected by UNESP Press and nominated for the 64th Jabuti Prize by Brazilian Book Chamber.



WEVERSON S. PEREIRA received the B.Sc. degree in computer science from the Centro Universitário da FEI, São Bernardo do Campo, Brazil, in 2022, where he is currently pursuing the M.Sc. degree in electrical engineering with an emphasis on signal and image processing. His research interests include machine learning, computer vision, natural language processing, and generative artificial intelligence.



PAULO S. S. SILVA received the bachelor's degree in computer science from the Federal University of Pará, Belém, Brazil, in 1996, the master's and Ph.D. degrees in computer science from the Federal University of Minas Gerais, Belo Horizonte, Brazil, in 1999 and 2003, respectively, and the Postdoctoral degree from the National Laboratory for Scientific Computing, in 2006.

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FERNANDO P. RIVERA received the bachelor's degree in electronic engineering from the National University of Engineering (UNI), Lima, Peru, in 2005, and the master's and Ph.D. degrees in electrical engineering from the University of Campinas (UNICAMP), Brazil, in 2011 and 2014, respectively.